Question #1 of 37

Question ID: 1473935

Diversification can reduce:

A) unsystematic risk.

B) macroeconomic risks.

C) systematic risk.

Question #2 of 37

Question ID: 1473926

Which of the following is NOT an assumption necessary to derive the arbitrage pricing theory (APT)?

A) The priced factors risks can be hedged without taking short positions in any portfolios.

B) Asset returns are described by a k-factor model.

C) A large number of assets are available to investors.

Question #3 of 37

Question ID: 1473947

Assume you are considering forming a common stock portfolio consisting of 25% Stonebrook Corporation (Stone) and 75% Rockway Corporation (Rock). As expressed in the two-factor returns models presented below, both of these stocks' returns are affected by two common factors: surprises in interest rates and surprises in the unemployment rate.

$$R_{Stone} = 0.11 + 1.0F_{Int} + 1.2F_{Un} + \varepsilon_{Stone}$$

$$R_{Rock} = 0.13 + 0.8F_{Int} + 3.5F_{Un} + \varepsilon_{Rock}$$

Assume that at the beginning of the year, interest rates were expected to be 5.1% and unemployment was expected to be 6.8%. Further, assume that at the end of the year, interest rates were actually 5.3%, the actual unemployment rate was 7.2%, and there were no company-specific surprises in returns. This information is summarized in Table 1 below:

Table 1: Expected versus Actual Interest Rates and Unemployment Rates

	Actual	**Expected**	**Company-specific returns surprises**
Interest Rate	0.053	0.051	0.0
Unemployment Rate	0.072	0.068	0.0

What is the predicted return for Stonebrook if the return unexplained by the model was -1%?

A) 12.00%

B) 10.68%.

C) 1.40%.

Question #4 of 37

Question ID: 1473933

Marcie Deiner is an investment manager with G&G Investment Corporation. She works with a variety of clients who differ in terms of experience, risk aversion and wealth. Deiner recently attended a seminar on multifactor analysis. Among other things, the seminar taught how the assumptions concerning the Arbitrage Pricing Theory (APT) model are different from those of the Capital Asset Pricing Model (CAPM). One of the examples used in the seminar is below.

$E(R_i) = R_f + f1\ B_{i,1} + f_2\ B_{i,2} + f_3\ B_{i,3}$. where: f_1 =3.0%, f_2 = −40.0%, and f_3 =50.0%.

Beta estimates for Growth and Value funds for a three factor model			
	Factor 1	**Factor 2**	**Factor 3**
Betas for Growth	0.5	0.7	1.2
Betas for Value	0.2	1.8	0.6

For the model used as an example in the seminar, if the T-bill rate is 3.5%, what are the expected returns for the Growth and Value Funds?

	$E(R_{Growth})$	$E(R_{Value})$
A)	3.1%	−3.16%
B)	33.5%	−41.4%
C)	37.0%	−37.9%

Question #5 of 37

Question ID: 1473930

The Arbitrage Pricing Theory (APT) has all of the following characteristics *EXCEPT* it:

A) assumes that asset returns are described by a factor model.

B) is an equilibrium pricing model.

C) assumes that arbitrage opportunities are available to investors.

Question #6 of 37

Question ID: 1473928

Which of the following is not an assumption of the arbitrage pricing theory (APT)?

A) The market contains enough stocks so that unsystematic risk can be diversified away.

B) Returns on assets can be described by a multi-factor process.

C) Security returns are normally distributed.

Question #7 of 37

Question ID: 1473950

A tracking portfolio is a portfolio with:

A) factor sensitivities of zero to all factors, positive expected net cash flow, and an initial investment of zero.

B) a factor sensitivity of one to a particular factor in a multi-factor model and zero to all other factors.

C) a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index.

Question #8 of 37

Question ID: 1473939

Assume you are attempting to estimate the equilibrium expected return for a portfolio using a two-factor arbitrage pricing theory (APT) model. Assume that you have estimated the risk premium for factor 1 to be 0.02, and the risk premium for factor 2 to be 0.03. The sensitivity of the portfolio to factor 1 is –1.2 and the portfolios sensitivity to factor 2 is 0.80. Given a risk free rate equal to 0.03, what is the expected return for the asset?

A) 5.0%.

B) 3.0%.

C) 2.4%.

Question #9 of 37

Question ID: 1473953

Janice Barefoot, CFA, has been managing a portfolio for a client who has asked Barefoot to use the Dow Jones Industrial Average (DJIA) as a benchmark. In her first year Barefoot managed the portfolio by choosing 29 of the 30 DJIA stocks. She selected a non-DJIA stock in the same industry as the omitted stock to replace that stock. Compared to the DJIA, Barefoot has placed a higher weight on the financial stocks and a lower weight on the other stocks still in the portfolio. Over that year, the non-DJIA stock in the portfolio had a negative return while the omitted DJIA stock had a positive return. The portfolio managed by Barefoot outperformed the DJIA. Based on this we can say that the return from factor tilts and asset selection were:

A) both positive.

B) positive and negative respectively.

C) negative and positive respectively.

Question #10 of 37

Question ID: 1473943

Identify the *most* accurate statement regarding multifactor models from among the following.

A) Macroeconomic factor models include explanatory variables such as real GDP growth and the price-to-earnings ratio and fundamental factor models include explanatory variables such as firm size and unexpected inflation.

B) Macroeconomic factor models include explanatory variables such as the business cycle, interest rates, and inflation, and fundamental factor models include explanatory variables such as firm size and the price-to-earnings ratio.

C) Macroeconomic factor models include explanatory variables such as firm size and the price-to-earnings ratio and fundamental factor models include explanatory variables such as real GDP growth and unexpected inflation.

Question #11 of 37

Question ID: 1473931

Which of the following does NOT describe the arbitrage pricing theory (APT)?

A) It is an equilibrium-pricing model like the CAPM.

B) It requires a weaker set of assumptions than the CAPM to derive.

C) There are assumed to be at least five factors that explain asset returns.

Question #12 of 37

Question ID: 1473932

One of the assumptions of the arbitrage pricing theory (APT) is that there are no arbitrage opportunities available. An arbitrage opportunity is:

A) a factor portfolio with a positive expected risk premium.

B) an investment that has an expected positive net cash flow but requires no initial investment.

C) a portfolio with factor exposures that sum to one.

Marianne Belair, CFA, is a wealth manager for a well-known company in Paris, France. She has developed macroeconomic factor models on portfolios Alpha and Bravo.

Equations for the two portfolios:

$$R_{Alpha} = 0.08 – 0.7\ F_{INFL} + 1.2\ F_{GDP}$$

$$R_{Bravo} = 0.13 + 0.6\ F_{INFL} + 2.3\ F_{GDP}$$

Belair has asked her colleague Pierre Louboutin to calculate the return attributable to a 1.5% surprise in GDP for an equally weighted portfolio comprising Alpha and Bravo.

Meanwhile, Belair is looking at Merci, a beauty stock for which she has developed a macroeconomic factor model. The arbitrage-pricing model shows a required return of 10% and the company-specific surprise for the year was 2%. Exhibit 1 shows additional information on the model:

Exhibit 1:

Variable	Actual Value (%)	Expected Value (%)	Factor Sensitivity
Interest rates	3.5%	2.5%	–0.3
Unemployment level	6.5%	5.5%	–0.7

Emily Grant, a senior manager at the firm, asks Louboutin to analyze the performance of three managers using the information in Exhibit 2.

Exhibit 2: Decomposing Active Risk

Portfolio	Active factor risk squared (%)	Active specific risk squared (%)	Active risk squared (%)	Active factor risk (% of Total Active Risk)	Active specific risk (% of Total Active Risk)	Active risk (%)
EM	0.5	0.5	1	50	50	1
EC	25.2	10.8	36	70	30	6
EV	21.6	14.4	36	60	40	6

Finally, Belair would like to capitalize on her expectation that real business activity will increase over the next year. As a separate concern, she has some existing positive exposure to inflation risk, which she would like to hedge. To achieve her goals she can use the portfolios in the Exhibit 3 which show the five relevant factors and respective factor sensitivities:

Exhibit 3:

Risk Factor	A	B	C	D	E
Confidence	0.10	1.00	0.00	0.70	0.00
Time horizon	0.00	0.00	0.00	0.50	0.00
Inflation	1.00	0.00	0.00	0.30	1.00
Business cycle	0.90	1.00	1.00	0.00	0.00
Market timing	1.00	0.00	0.00	0.90	0.00

Question #13 - 16 of 37

Question ID: 1473957

Pierre's answer to Belair's first request regarding the equally weighted portfolio, is *closest* to:

A) 1.75%.

B) 2.13%.

C) 2.63%.

Question #14 - 16 of 37

Question ID: 1473958

The actual return of Merci is *closest* to:

A) 9%.

B) 10%.

C) 11%.

Question #15 - 16 of 37

Question ID: 1473959

Using Exhibit 2, the portfolio that has the *most* exposure to asset selection risk is:

A) EM.

B) EC.

C) EV.

Question #16 - 16 of 37

Question ID: 1473960

Which two portfolios from Exhibit 3 best achieve Belair's goals in relation to business activity and inflation risk?

A) B and A.

B) B and E.

C) C and E.

Question #17 of 37

Question ID: 1473941

A multi-factor model that uses unexpected changes (surprises) in macroeconomic variables (e.g., inflation and gross domestic product) as the factors to explain asset returns is called a:

A) macroeconomic factor model.

B) fundamental factor model.

C) statistical factor model.

Question #18 of 37

Question ID: 1473961

A portfolio manager uses a two-factor model to manage her portfolio. The two factors are confidence risk and time-horizon risk. If she wants to bet on an unexpected increase in the confidence risk factor (which has a positive risk premium), but hedge away her exposure to time-horizon risk (which has a negative risk premium), she should create a portfolio with a sensitivity of:

A) −1.0 to the confidence risk factor and 1.0 to the time-horizon factor.

B) 1.0 to the confidence risk factor and -1.0 to the time-horizon factor.

C) 1.0 to the confidence risk factor and 0.0 to the time-horizon factor.

Question #19 of 37

Question ID: 1473936

Michael Paul, a portfolio manager, is screening potential investments and suspects that an arbitrage opportunity may be available. The three portfolios that meet his screening criteria are detailed below:

Portfolio	Expected Return	Beta
X	12%	1.0
Y	16%	1.3
Z	8%	0.9

Which of the following portfolio combinations produces the highest return while maintaining a beta of 1.00?

	Portfolio X	Portfolio Y	Portfolio Z
A)	25%	50%	25%
B)	50%	12%	38%
C)	100%	0%	0%

Question #20 of 37

Question ID: 1473940

Portfolios A and B have an expected return of 4.4% and 5.3% respectively. Assume that a one-factor APT model is appropriate and the factor sensitivities of portfolios A and B are 0.8 and 1.1 respectively. The risk-free rate and factor risk premium are *closest* to:

	Risk Free Rate	Factor Risk Premium
A)	3.00%	2.00%
B)	2.50%	3.00%
C)	2.00%	3.00%

Question #21 of 37

Question ID: 1473951

A common strategy in bond portfolio management is *enhanced indexing by matching primary risk factors*. This strategy could be implemented by forming:

A) a portfolio with factor sensitivities that sum to one.

B) a portfolio with factor sensitivities equal to that of the index.

C) a portfolio with asset portfolio weights equal to that of the index.

Question #22 of 37

Question ID: 1473944

Summer Vista decides to develop a fundamental factor model. She establishes a proxy for the market portfolio, and then considers the importance of various factors in determining stock returns. She decides to use the following factors in her model:

- Changes in payout ratios.
- Credit rating changes.
- Companies' position in the business cycle.
- Management tenure and qualifications.

Which of the following factors is *least appropriate* for Vista's factor model?

A) Management tenure and qualifications.

B) Changes in payout ratios.

C) Companies’ position in the business cycle.

Question #23 of 37

Question ID: 1473949

A portfolio with a factor sensitivity of one to a particular factor in a multi-factor model and zero to all other factors is called a(n):

A) tracking portfolio.

B) factor portfolio.

C) arbitrage portfolio.

Question #24 of 37

Question ID: 1473934

Arbitrage pricing models assume which risk is priced?

A) Systematic.

B) Both systematic and unsystematic.

C) Unsystematic.

Question #25 of 37

Question ID: 1473963

Janice Barefoot, CFA, has managed a portfolio where she used the Dow Jones Industrial Average (DJIA) as a benchmark. In the past two years the average monthly return on her portfolio has been higher than that of the DJIA. To get a measure of active return per unit of active risk Barefoot should compute the:

A) Sharpe ratio, which is the standard deviation of the differences between the portfolio and benchmark returns divided into the average of those differences.

B) information ratio, which is the average excess portfolio return over the benchmark divided by the standard deviation of the differences between the portfolio and benchmark returns.

C) information ratio, which is the standard deviation of the differences between the portfolio and benchmark returns divided by the average of those differences.

Question #26 of 37

Question ID: 1473942

The macroeconomic factor models for the returns on Omni, Inc., (OM) and Garbo Manufacturing (GAR) are:

$$R_{OM} = 20.0\% + 1.0(F_{GDP}) + 1.4(F_{QS}) + \varepsilon_{OM}$$

$$R_{GAR} = 15.0\% + 0.5(F_{GDP}) + 0.8\ (F_{QS}) + \varepsilon_{GAR}$$

What is the expected return on a portfolio invested 60% in Omni and 40% in Garbo?

A) 18.0%.

B) 20.96%.

C) 19.96%.

Question #27 of 37

Question ID: 1473927

Which of the following is an equilibrium-pricing model?

A) The arbitrage pricing theory (APT).

B) Fundamental factor model.

C) Macroeconomic factor model.

Question #28 of 37

Question ID: 1473937

Given a three-factor arbitrage pricing theory (APT) model, what is the expected return on the Premium Dividend Yield Fund?

- The factor risk premiums to factors 1, 2 and 3 are 8%, 12% and 5%, respectively.
- The fund has sensitivities to the factors 1, 2, and 3 of 2.0, 1.0 and 1.0, respectively.
- The risk-free rate is 3.0%.

A) 33.0%.

B) 36.0%.

C) 50.0%.

Question #29 of 37

Question ID: 1473954

Janice Barefoot, CFA, has been managing a portfolio for a client who has asked Barefoot to use the Dow Jones Industrial Average (DJIA) as a benchmark. In her second year, Barefoot used 29 of the 30 DJIA stocks. She selected a non-DJIA stock in the same industry as the omitted DJIA stock to replace that stock. Compared to the DJIA, Barefoot placed a lower weight on the communication stocks and a higher weight on the other stocks still in the portfolio. Over that year, the non-DJIA stock in the portfolio had a positive and higher return than the omitted DJIA stock. The communication stocks had a negative return while all of the other stocks had a positive return. The portfolio managed by Barefoot outperformed the DJIA. Based on this we can say that the return from factor tilts and asset selection were:

A) both positive.

B) negative and positive respectively.

C) positive and negative respectively.

Question #30 of 37

Question ID: 1473955

Rob Tanner, portfolio manager at Alpha Inc. meets his old college friend Del Torres for lunch. Torres excitedly tells Tanner about his latest work with tracking and factor portfolios. Torres says he has developed a tracking portfolio to aid in speculating on oil prices and is working on a factor portfolio with a specific set of factor sensitivities to the Russell 2000.

Did Torres correctly describe tracking and factor portfolios?

	Tracking	Factor
A)	Yes	No
B)	No	Yes
C)	No	No

Question #31 of 37

Question ID: 1473938

Given a three-factor arbitrage pricing theory APT model, what is the expected return on the Freedom Fund?

- The factor risk premiums to factors 1, 2, and 3 are 10%, 7% and 6%, respectively.
- The Freedom Fund has sensitivities to the factors 1, 2, and 3 of 1.0, 2.0 and 0.0, respectively.
- The risk-free rate is 6.0%.

A) 24.0%.

B) 30.0%.

C) 33.0%.

Question #32 of 37

Question ID: 1473962

Question #32 of 37

Question ID: 1473962

In the context of multi-factor models, investors with lower-than-average exposure to recession risk (e.g. those without labor income) can earn a risk premium for holding dimensions of risk unrelated to market movements by creating equity portfolios with:

A) greater-than-average market risk exposure.

B) greater-than-average exposure to the recession risk factor.

C) less-than-average exposure to the recession risk factor.

Question #33 of 37

Question ID: 1473946

Assume you are considering forming a common stock portfolio consisting of 25% Stonebrook Corporation (Stone) and 75% Rockway Corporation (Rock). As expressed in the two-factor returns models presented below, both of these stocks' returns are affected by two common factors: surprises in interest rates and surprises in the unemployment rate.

$$R_{Stone} = 0.11 + 1.0F_{Int} + 1.2F_{Un} + \varepsilon_{Stone}$$

$$R_{Rock} = 0.13 + 0.8F_{Int} + 3.5F_{Un} + \varepsilon_{Rock}$$

Assume that at the beginning of the year, interest rates were expected to be 5.1% and unemployment was expected to be 6.8%. Further, assume that at the end of the year, interest rates were actually 5.3%, the actual unemployment rate was 7.2%, and there were no company-specific surprises in returns. This information is summarized in Table 1 below:

Table 1: Expected versus Actual Interest Rates and Unemployment Rates

	Actual	Expected	Company-specific returns surprises
Interest Rate	0.053	0.051	0.0
Unemployment Rate	0.072	0.068	0.0

What is the portfolio's sensitivity to interest rate surprises?

A) 0.25.

B) 0.85.

C) 0.95.

Question #34 of 37

Question ID: 1473952

The Real Value Fund is designed to have zero exposure to inflation. However its current inflation factor sensitivity is 0.30. To correct for this, the portfolio manager should take a:

A) 30% short position in the inflation tracking portfolio.

B) 30% long position in the inflation factor portfolio.

C) 30% short position in the inflation factor portfolio.

Question #35 of 37

Question ID: 1473929

Which of the following is NOT an underlying assumption of the arbitrage pricing theory (APT)?

A) A market portfolio exists that contains all risky assets and is mean-variance efficient.

B) Asset returns are described by a K factor model.

C) There are a sufficient number of assets for investors to create diversified portfolios in which firm-specific risk is eliminated.

Question #36 of 37

Question ID: 1473945

Assume you are considering forming a common stock portfolio consisting of 25% Stonebrook Corporation (Stone) and 75% Rockway Corporation (Rock). As expressed in the two-factor returns models presented below, both of these stocks' returns are affected by two common factors: surprises in interest rates and surprises in the unemployment rate.

$$R_{Stone} = 0.11 + 1.0F_{Int} + 1.2F_{Un} + \varepsilon_{Stone}$$

$$R_{Rock} = 0.13 + 0.8F_{Int} + 3.5F_{Un} + \varepsilon_{Rock}$$

Assume that at the beginning of the year, interest rates were expected to be 5.1% and unemployment was expected to be 6.8%. Further, assume that at the end of the year, interest rates were actually 5.3%, the actual unemployment rate was 7.2%, and there were no company-specific surprises in returns. This information is summarized in Table 1 below:

Table 1: Expected versus Actual Interest Rates and Unemployment Rates

	Actual	Expected	Company-specific returns surprises
Interest Rate	0.053	0.051	0.0
Unemployment Rate	0.072	0.068	0.0

What is the expected return for Stonebrook in the absence of surprises?

A) 11.0%.

B) 13.2%.

C) 13.0%.

Question #37 of 37

Question ID: 1473948

A portfolio with a specific set of factor sensitivities designed to replicate the factor exposures of a benchmark index is called a:

A) arbitrage portfolio.

B) tracking portfolio.

C) factor portfolio.